Cost Optimization

Using the Greedy Heuristic Method, we achieved an average coverage rate of 98% over one year with 208 cellular antennas. With the Genetic Algorithm, we increased the average coverage rate to 99.8% while reducing the number of antennas to 187.
The coverage rate is now close to perfect.

Cost Model Assumptions

- CapEx (Capital Expenditure) → One-time cost to buy and deploy equipment. Includes: antenna hardware, installation (labor, permits, cranes, cabling), and initial testing.
    - Assumed value: €12,000 per antenna
      (Based on industry ranges of €9k–€46k per small-cell site deployment, depending on location and complexity) [1]
- OpEx (Operational Expenditure) → Recurring cost to keep the system running. Includes: site lease/rent (if private), electricity, maintenance & repairs, monitoring, insurance.
    - Assumed value: €1,500 per antenna per year
      (Derived from conservative estimates of light maintenance and small-site operational costs, below the upper bound of €600–€12,000/year seen in industry sources [2]

Savings with Genetic Algorithm

- **CapEx Savings:**
  (208 – 187) × €12,000 = **€252,000** (one-time)
- **OpEx Savings:**
  (208 – 187) × €1,500 = **€31,500 per year**

Summary

- 21 fewer antennas required with Genetic Algorithm
- Coverage improved: 98% → 99.8%
- Cost reduction: ~€252k upfront + ~€31.5k saved annually
- Conclusion: Genetic Algorithm achieves better coverage with lower infrastructure and operating costs.

References

[6] PatentPC, *5G Infrastructure Costs: What Telcos Are Paying*.

Available: https://patentpc.com/blog/5g-infrastructure-costs-what-telcos-are-paying/

[7] Steel In The Air, *Small Cell Towers and Leases*.

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